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Routine 2D inversion of magnetotelluric data using the determinant of the impedance tensor

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ABSTRACT

Recent developments in the speed and quality of data acquisition using the radiomagnetotelluric (RMT) method, whereby large amounts of broadband RMT data can be collected along profiles, have prompted us to develop a strategy for routine inverse modeling using 2D models. We build a rather complicated numerical model containing both 2D and 3D elements believed to be representative for shallow conductors in crystalline basement overlain by a thin sedimentary cover. We then invert the corresponding synthetic data on selected profiles, using both traditional MT approaches, as well as the proposed approach, which is based on the determinant of the MT impedance tensor. We compare the estimated resistivity models with the true models along the selected profiles and find that the traditional approaches often lead to strongly biased models and bad data fit, in contrast to those using the determinant. In this case, much of the bias is removed and the data fit is improved. The determinant of the impedance tensor is independent of the chosen strike direction, and once the a priori model is set, the best fitting model is found to be practically independent of the starting model used. We conclude that the determinant of the impedance tensor is a useful tool for routine inverse modeling.

INTRODUCTION

Since the early days of the magnetotelluric (MT) method, it has been customary to interpret impedance and tipper data by considering only part of the information to reasonably construct 1D or 2D models believed to represent the real 3D earth structure. Only in rare cases do MT data obey strict 1D or 2D conditions. For 1D models, it has been customary to use some scalar property of the impedance tensor to be modeled, for example, the difference between the off-diagonal elements or the determinant (Park and Livelybrooks, 1989; Agarwal et al., 1993; Mauriello et al., 2000; Berdichevsky, 1999; Bai et al., 2001; Ingham et al., 2001). For 2D models, the trend has rather been to model both modes (the transverse electric or TE mode and the transverse magnetic or TM mode) and try to get the best out of them by various ad hoc selections of data. Berdichevsky et al. (1998) argue that both TE and TM modes should be used at various stages in the interpretation process. This is in contrast to many scientists in the western hemisphere (Wannamaker et al., 1984; Boerner et al., 1999) who prefer simply to use the TM mode, because it is generally believed to be less influenced by 3D effects than the TE mode, although Berdichevsky et al. (1998) argue that this is true only for 3D conducting structures, whereas for 3D resistive structures, the TE mode is less distorted.

In parallel with modeling studies, much effort has been made to decompose impedance tensors into local and regional parts (Bahr, 1988, 1991; Zhang et al., 1987; Groom and Bailey, 1991) with the idea that the regional structure lends itself to simpler analysis with either 1D or 2D models. In some cases, the concept of a regional 2D model has undoubtedly been successful, but in general, the dimensionality of the regional impedance tensor is neither 1D or 2D, but 3D when looked upon over a wide frequency range. This means that it may become difficult to model the regional impedance tensor with simultaneous use of TE and TM modes because of mode mixing as the frequency varies.

Another development for imaging 3D structures directly from the data is the use of invariants of the impedance tensor [see Weaver et al. (2000) for a general description]. Szarka and Menvielle (1997), in their review of the seven rotational invariants of the impedance tensor, conclude, on the basis of some simple thin-sheet-model calculations, that certain invariants, among them the determinant, provide a good basis for studying the 3D morphology of resistivity structures. Caldwell and Bibby (1998) arrived at a similar conclusion for time-domain LOTEM measurements. The instantaneously apparent resistivity tensor can be used to image main features of subsurface resistivity variations. Maps

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of the three independent tensor invariants provide simple "time slices" through rather complicated structures that can be clearly recognized in the maps.

In this paper, we propose to directly invert the determinant invariant of the MT impedance tensor for a suite of MT stations along a profile using a 2D model. The measured determinant invariant is modeled by the corresponding 2D invariant, which—except for a change of sign—is the product of the TE and TM impedances. We study the effectiveness of 2D inversion using a 3D model describing typical variations of electrical resistivity in the shallow crust. We first describe the 3D model and the corresponding synthetic data before we show results of 2D inversion along a few profiles across the 3D model.

TRANSFER FUNCTIONS

In the ideal case, a magnetotelluric survey consists of a number of stations adequately distributed in the area of study, but in practice, MT data are often collected along profiles or along crooked lines that are projected onto a hypothetical survey line, along which 2D modeling is carried out.

At each station, three magnetic and two horizontal electric components are measured, from which the impedance tensor and tipper vector are estimated, together with their standard errors. Modern robust data-processing techniques extracting the transfer functions from the measured time series are generally believed to provide estimates of transfer functions and estimates of their errors that reasonably well describe the deviation of the estimates away from their true values. Nevertheless, bias effects in the data give rise to inconsistencies that can be caused by both industrial disturbances, such as railways and DC-AC transformers, as well as by natural disturbances, for which the fundamental assumption of a plane source no longer holds. Whereas the former disturbance is particularly important in the frequency band 500 kHz to 100 s, the latter is the dominant factor for longer periods, say from a few hundred seconds and upwards, where the lower limit depends strongly on the electrical conductivity structure in the area.

In the case of 2D structures the magnetotelluric impedance tensor

$$\mathbf{Z} = \begin{Bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{Bmatrix}, \tag{1}$$

connecting horizontal components of the surface electromagnetic field at a particular position and frequency, becomes particularly simple in the principal coordinate system, with the x-axis pointing in the strike direction:

$$\mathbf{E}_h = \begin{Bmatrix} E_x \\ E_y \end{Bmatrix} = \begin{Bmatrix} 0 & Z_{TE} \\ Z_{TM} & 0 \end{Bmatrix} \begin{Bmatrix} H_x \\ H_y \end{Bmatrix} = \mathbf{Z}_{2D}\mathbf{H}_h, \tag{2}$$

where subscripts TE and TM stand for transverse electric and transverse magnetic modes, respectively; i.e., currents flow parallel or perpendicular to the structure, and

$$\mathbf{E}_h = \begin{Bmatrix} E_x \\ E_y \end{Bmatrix} \quad \text{and} \quad \mathbf{H}_h = \begin{Bmatrix} H_x \\ H_y \end{Bmatrix} \tag{3}$$

denote horizontal electric and magnetic fields, respectively.

The determinant of the impedance tensor is rotationally invariant, and the determinant impedance for a 2D model can therefore, in general, be written as

$$\begin{aligned} Z_{DET} &= |Z_{DET}| \exp[i\phi_{DET}] = \sqrt{Z_{TE}(-Z_{TM})} \\ &= \sqrt{|Z_{TE}||Z_{TM}|} \exp\left[\tfrac{1}{2} i(\phi_{TE} + \phi_{TM})\right], \end{aligned} \tag{4}$$

where the TM phase is defined from the negative of the TM impedance to get positive phases for a 1D environment [using an $\exp(i\omega t)$ time dependence, as in Weidelt, 1972]. Transforming to logarithmic variables, i.e., the logarithm of apparent resistivity and the phase, directly gives

$$\begin{aligned} \log \rho_{DET}^{app} &= \tfrac{1}{2}\left(\log \rho_{TE}^{app} + \log \rho_{TM}^{app}\right) \quad \text{and} \\ \phi_{DET} &= \tfrac{1}{2}(\phi_{TE} + \phi_{TM}), \end{aligned} \tag{5}$$

whereby 2D determinant data can be considered as the arithmetic mean of TE and TM mode data.

When dealing with real data (or synthetic data from 3D structures), we use the full impedance tensor to calculate the determinant impedance, but we model it by the 2D determinant expressed in equation 2.

ONE-DIMENSIONAL MODELS

Strictly speaking, 1D models explain the data only if their responses fit the measured data within their uncertainties. With today's data quality, this rarely happens, because field data nearly always show 3D effects in the form of nonvanishing tipper vectors and diagonal impedance elements, as well as off-diagonal elements not adding to zero.

When data from a particular station is interpreted with a 1D model, there are many possibilities for combining the 8 degrees of freedom contained in the impedance tensor at each frequency into 2 degrees that can be fitted with a 1D model; for example, an apparent resistivity and a phase as functions of frequency (or period). Typically, one of the off-diagonal elements or two rotational invariants of the impedance are selected. If the study area can reasonably be modeled in 1D, as is often the case in near-surface studies, all such models will normally show the same general trends in electrical resistivity distribution in the area, both laterally and with depth. In such cases, therefore, it can be argued that the determinant of the impedance should be chosen because it not only gives a logarithmic average of the two principal off-diagonal impedance elements, but also mimics the phase correctly if the regional structure is 1D or 2D.

Generally, the better the fit between measured data and predicted data, the better the model resolution. One-dimensional model fits are generally better than 2D model fits, partly because the two-model space may not be as well sampled as the corresponding 1D model space, but primarily because 2D models must satisfy data at all stations. Thus, effects of bad data quality or 3D effects may swamp resolution. Of course, the finer details that can be resolved by 1D models should be seen on a suite of stations to be able to say with confidence that they represent true depth variations in conductivity and that they are not dominated by side-effects caused by 2D or 3D structures. Here, the information contained in the tipper vector can be of help to rule out or support the existence of side effects.

TWO-DIMENSIONAL MODELS

Two-dimensional modeling results may be difficult to explain to people outside the EM community, because three different models representing best fit to TE, TM, and TE+TM modes, respectively, are often shown. The two different current modes enhance lateral conductivity variations differently. Generally, the TE mode couples well to conductors, whereas the TM mode provides a better coupling to resistors. This is illustrated by the example of Figure 1, where the synthetic data (50 stations with a spacing of 10 m with frequencies 4, 8, 16, 32, 64, 128, and 256 kHz) from isolated conductors and resistors were inverted using an a priori model of 400 ohm-m. All models have an rms data fit of 1%.

The TE data couple nicely to the conductor to the left (Figure 1b), although some smoothing is obvious, and the resistivity only reaches a level of about 100 ohm-m compared with the true value of 40 ohm-m. The resistor to the right is hardly visible. Its resistivity is less than 500 ohm-m, only slightly higher than the background of 400 ohm-m.

The TM data (Figure 1c) couple much better to the resistor to the right, although some smoothing is obvious, and the resistivity only reaches a level of about 600 ohm-m compared with the true value of 4000 ohm-m. The conductor to the left is hardly visible. Its resistivity is higher than 250 ohm-m, only slightly lower than the background resistivity.

The DET data (Figure 1e) provide a model that can be viewed as some mean of the TE and TM models. The conductor and the resistor are clearly resolved, but not so well as for TE and TM data alone, respectively.

The best result, obviously, is found when combining both TE and TM modes (Figure 1d). The conductor to the left has a minimum resistivity of only 60 ohm-m, and the resistor to the right has a maximum resistivity of 850 ohm-m.

In standard 2D modeling, an attempt is often made to fit TE and TM data independently, whereby two different best-fitting models are generally produced. This is partly caused by the fact, as illustrated above, that TE and TM modes respond differently to specific conductivity models and partly because of inconsistencies between the 2D-model world and the real 3D world. The latter effect can, to some extent, be accommodated in the TM mode by regularization approaches to the inverse problem; that is, allowing the upper part of the model to be somewhat rougher than the remaining deeper part. The result is that the upper part may incorporate effects of static shifts without specifically defining them as part of the inverse problem. The TE mode, in contrast, consists of a purely inductive response with no electric charges and cannot be fitted in the same way. At best, an explicit static-shift parameter can be introduced to describe the same effect, because the near-surface lateral inhomogeneities in a 2D model hardly perturb the TE responses, at least at frequencies for which the inductive scale length is much larger than the scale of the inhomogeneities.

The complexity of the inverse problem is further underlined when an attempt is made to fit TE and TM modes simultaneously. If the real world can be well described by a regional 2D model, then simultaneous inversion of TE and TM modes can be quite successful if only the phase of the TE mode is used. However, in many cases, the simultaneous inversion of both modes leads to models lacking some of the sharpness (resolution) characterizing TM-mode inversion, because the 2D model is inconsistent with real 3D data, and the best-fitting model represents a bleak common-denominator model satisfying strict 2D constraints that are not found in the data.

MODEL STUDIES

The model studied in this section is related to near-surface problems, for which it is particularly important to search for a robust and simple approach because of large data volumes. However, it is not possible to cover all important model types. Only a model with a rather complicated 3D conducting structure simulating the conditions in a crystalline environment with an inhomogeneous overburden will be studied. Two-dimensional inversions were carried out with the Occam algorithm of Siripunvaraporn and Egbert (2000), slightly expanded with a few subroutines to enable inversion of the determinant in addition to the TM, TE, and TE+TM modes.

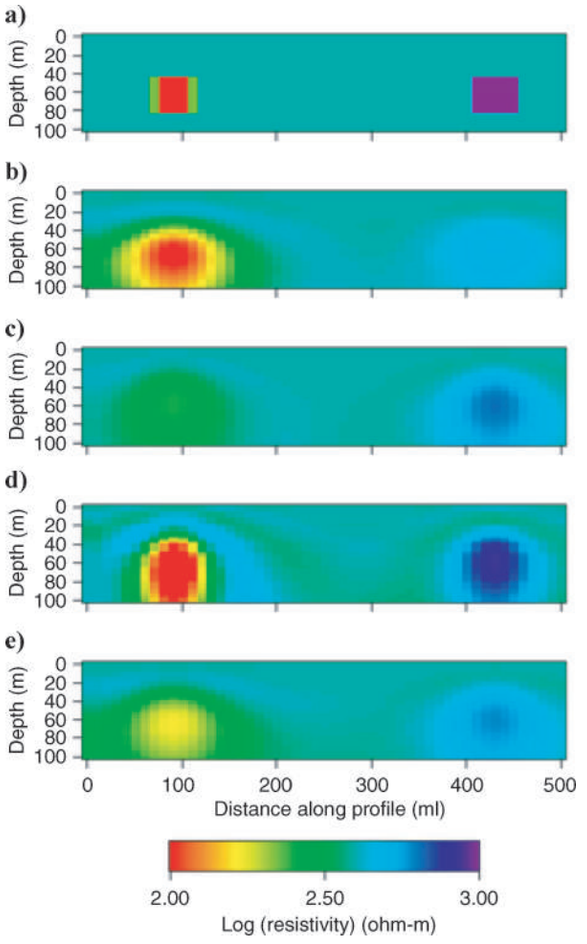


Figure 1. Test model and inverted models fitted to an rms prediction error of 1% of the corresponding impedances. (a) Test model of a homogeneous half-space with conductor of 100 ohm-m to the left and a resistor of 1000 ohm-m to the right. Inverted models using (b) TE mode alone, (c) TM mode alone, (d) TE+TM modes and (e) Determinant (DET) of the impedance tensor.

Shallow overburden and narrow conductor at depth: 3D data

The synthetic model is a combination of a 2D deeper structure and a 3D surface structure. There are two inhomogeneous layers, embedded in a normal background of 1,000 ohm-m for the upper 60 m and 10,000 ohm-m below. Both layers, sketched in Figure 2, have a thickness of 10 m and are at 0–10 m and 50–60 m depth. The top layer contains a two-loop 3D structure (formed like an eight) of 100 ohm-m, which results in a resistivity contrast of 10 against the background. The second layer describes an elongated conductor of 2 ohm-m, i.e., a resistivity contrast of 500 against the normal structure.

This model simulates a typical conductivity structure over crystalline basement: the loops in the surface layer represent alluvial fillings of minor "valleys" around small "hills," while the deep conductor images a small mineralized zone—all in a resistive background structure of moderately fractured crystalline rocks in the upper 60 m overlying more intact crystalline rocks below.

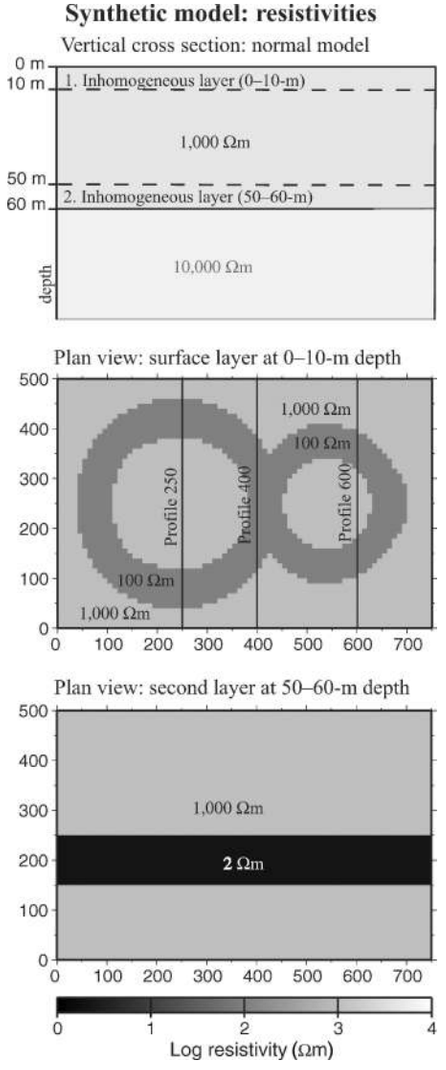


Figure 2. Vertical cross section and plan view of the three-dimensional model. The surface conductor is 10-m thick and has a resistivity of 100 ohm-m. Profiles 250, 400, and 600 have been selected for 2D inverse modeling. The conductor at 50-m depth is 10 m thick and has a resistivity of 2 ohm-m.

Modeling technique

Numerical modeling is performed by the powerful 3D code "X3D" (Avdeev, Kuvshinov, Pankratov et al., 2000; Avdeev et al., 2002). Electromagnetic responses are calculated by the integral equation technique, combining Krylov subspace iteration with the modified iterative dissipative method.

The total model area of 1750 m × 1100 m exceeds the area presented in Figure 2 (750 m × 500 m) by a frame of width 500 m and 300 m in the east-west and north-south directions, respectively. This enlarged model area helps to emphasize the 2D character of the deep elongated conductor. Square model cells with a base length of 5 m result in 350 × 220 model cells, embedded in the otherwise homogeneous normal structure. The vertical discretization of both layers varies between 2 m and 5 m.

Modeling frequencies are 1, 2, 4, 8, 16, 32, 64, and 128 kHz. Penetration depth for the latter is 14 m in the surface-layer conductor, being still large against the horizontal- and vertical-model cell dimensions. Penetration depth into the second layer is in the order of the cell-size dimension for this frequency range.

Modeling required about 30 iterations for a residual norm of 0.00001, resulting in a computation time of two minutes per iteration on a conventional PC. Note that this model is also suitable for the thin-sheet modeling technique by replacing both layers with thin sheets having identical conductance (product of conductivity and thickness) contrasts. Qualitatively similar results have been achieved using the multisheet code "XPLATES" of Avdeev, Kuvshinov, and Pankratov et al. (2000).

Model responses

Model responses for two frequencies, 128 and 2 kHz, are illustrated in Figures 3 and 4, respectively. Referring to the 2D-structure of the second layer with a conductor parallel to the abscissa, we define TE mode as associated with the electric-field direction parallel to the deep conductor (abscissa) and TM mode with the electric-field direction perpendicular to the deep conductor (ordinate). Apparent resistivity and impedance phase is presented for TE, TM, and Z_{DET} (DET) as well as real and imaginary induction arrows following the Wiese convention (real arrows pointing away from conductive structures). The synthetic data set will be used in the following section, Profile Inversion of Synthetic Data; however, the responses themselves already point out some basic characteristics of the different modes:

Figure 3 shows the model responses for 128 kHz.

1) Responses for shallow penetration depths image the loop-like surface layer structure; however, higher phases inside the loops already indicate the deep conductor.
2) TE and TM modes show oscillations at conductivity contrasts in apparent resistivity (TM-mode behavior) for loop segments perpendicular to the electric field (segments

parallel to the ordinate for TE mode and parallel to the abscissa for TM mode), while low phases inside conducting loops become pronounced when the electric field is tangential to the loops.

3) Z_{DET} provides the best image of the true structure for both the surface and the deeper conductor.
4) Real induction arrows image the surface conductor only, while imaginary induction arrows become vanishingly small.

Figure 4 shows the model responses for 2 kHz.

1) Responses for these penetration depths image the deep conductor, while the surface layer is still the dominant feature in apparent resistivities because of static shifts of the surface electric field.
2) High TE-mode phases clearly indicate the existence of the deep conductor, while TM-mode responses—although less pronounced—better locate the boundaries of the lower conductor (associated with charges). TE-mode phases even partly exceed 90° (out of quadrant) at locations where the surface loops and the deeper conductor superimposes at right angles.
3) Z_{DET} provides an accurate image of the deep structure, compromising between large response amplitudes and sharp boundary resolution.
4) Real induction arrows show a superposition of surface- and second-layer conductivity contrasts, while imaginary arrows more clearly point out the deep conductor, being less distorted by the surface structure. (Note the similarity to impedance phases, which are more sensitive to deeper structures than the corresponding apparent resistivities.)

These comparisons lead to the following findings:

1) Out-of-phase behavior has been observed in TE mode at 4, 2, and 1 kHz, where modes referring to different depth levels are "mixed": Either the TE mode for the deep structure (conductor parallel to abscissa) is accompanied by TM mode for the loop segment at the surface (conductor parallel to ordinate) or vice versa (the TM mode for the deep structure and TE mode for the surface structure). A simple mechanism to produce out-of-phase data has been found by superimposing two 2D structures at right angles to each other at slightly different depth levels. Such a model, substituting an elongated NS conductor for the surface loop structure, yielded out-of-phase behavior at 1 kHz, where the two conductors cross. At penetration depths that are sensitive to both conductors, TE- and TM-mode directions become mixed and may lead to phases out of quadrant.

The areas out of phase in TE mode increase considerably with decreasing frequencies and out-of-phase behavior, which has even been observed for Z_{DET} at 1 kHz. Model studies with various conductivity contrasts between the lower conductor and the background structure show the following dependence: An increased conductivity contrast increases the area for which out-of-quadrant effects occur, and such effects can be observed even at higher frequencies.

All the mode comparisons at various frequencies show that Z_{DET} has a more smooth response than TE or TM mode: Z_{DET} is more robust against 3D distortion.

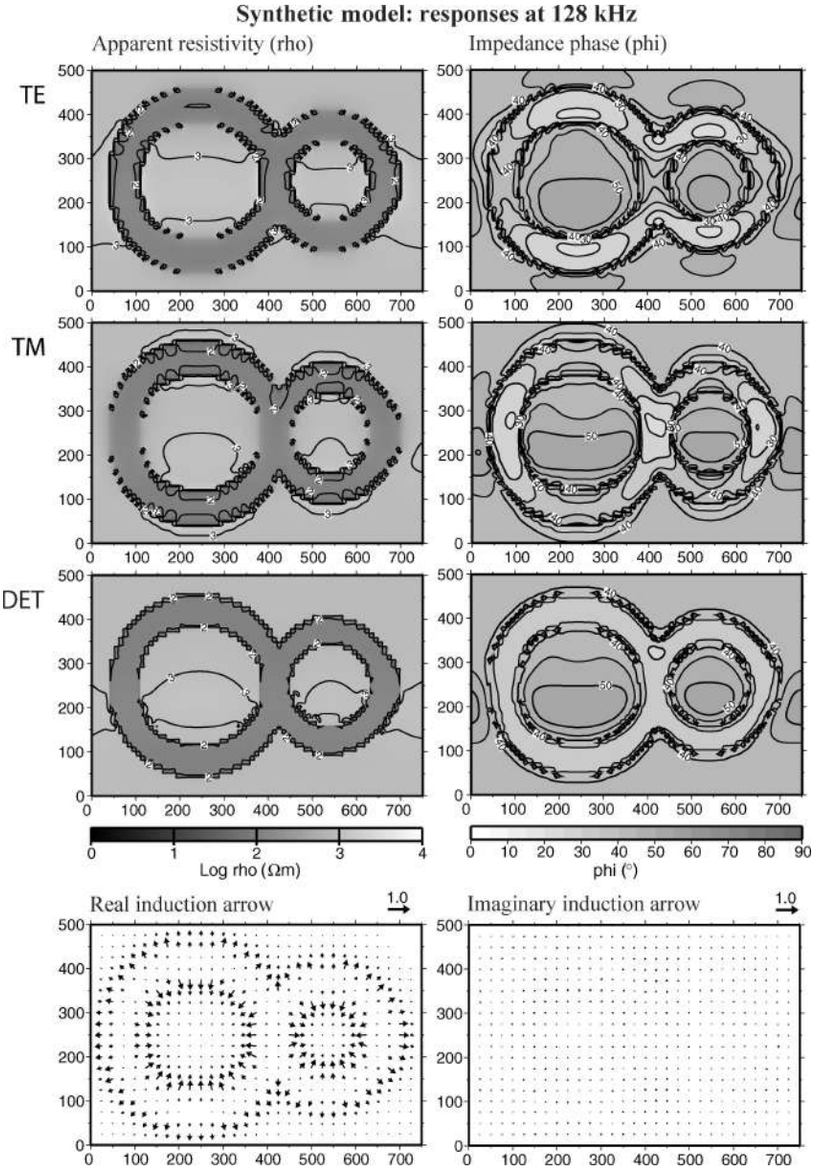


Figure 3. Model responses for 128 kHz, showing apparent resistivities and phases of off-diagonal elements and determinant of the impedance tensor and real and imaginary parts of the Wiese-vector. TM means primary electric field parallel to ordinate. TE means primary electric field parallel to abscissa.

2) Z_{DET} responses provide the best overall image, while TE- and TM-mode responses focus on specific segments of the loop structure (depending on the direction of the electric field).
3) Tipper data might provide useful additional information for inversion. For this model especially, the imaginary part is more sensitive to the deeper structure.

PROFILE INVERSION OF SYNTHETIC DATA

Plane-wave responses at the seven frequencies of 2, 4, 8, 16, 32, 64, and 128 kHz were taken as input data for the 2D inversion. No random noise was added to the responses, but a theoretical random error of 1% was assumed for the impedance, i.e., 2% for apparent resistivity and 0.57° for phase. The distance between measuring stations is 10 m, and the horizontal discretization of the model is taken to 5 m to allow the model to represent static-shift effects as small variations in the upper part of the model. The vertical discretization was taken to vary between 1 and 3 m in the upper 60 m of the model, gradually increasing to 512 m in a geometric progression to 900 m. In total, the number of model cells was 114 (horizontal) × 58 (vertical) = 6612 cells. On each profile of length 500 m, there are 51 stations. We found that the iterational process of finding solutions that best fit the data becomes stabilized by incorporating all data points for Fréchet derivative calculations. We studied the effect of varying the initial model and the a priori models. However, to keep the problem as simple and realistic as possible, we prefer to use identical initial and a priori models of homogeneous half-space types. In all models presented here, we used a 1000-ohm-m half-space.

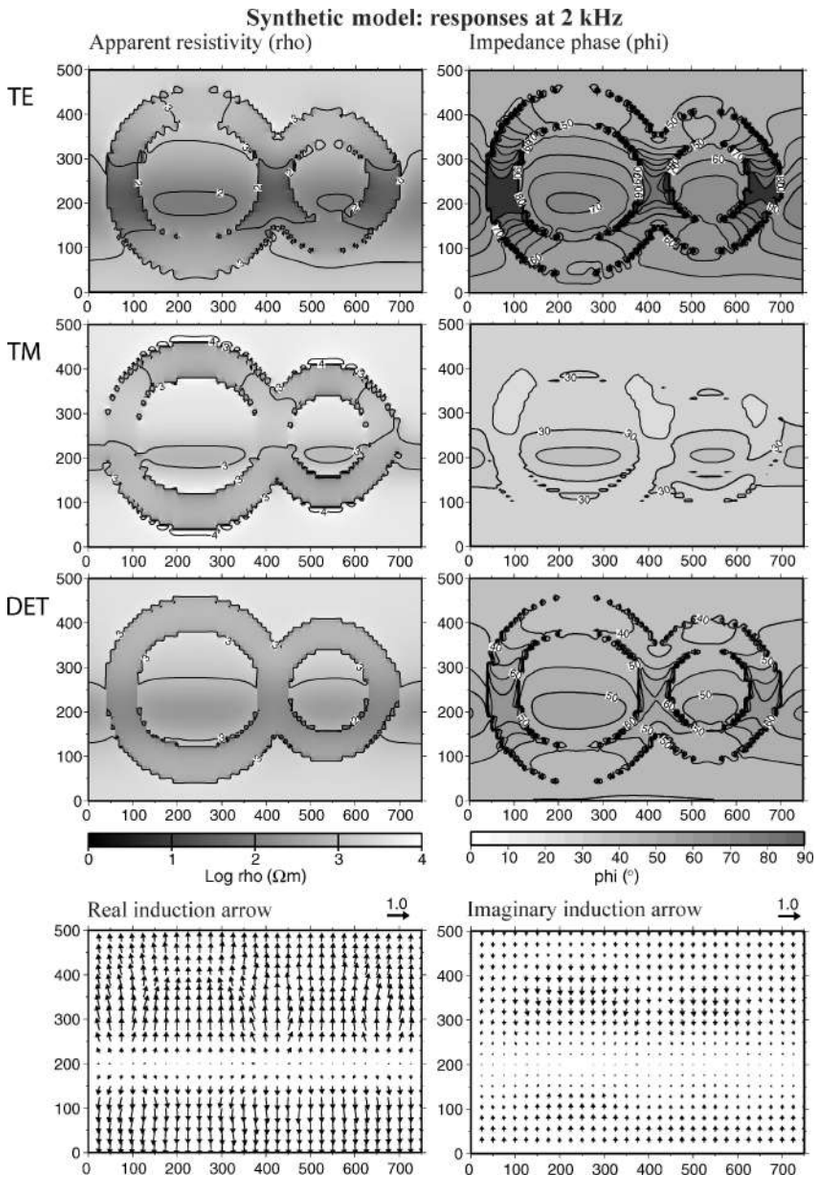


Figure 4. Model responses for 2 kHz, showing apparent resistivities and phases of off-diagonal elements and determinant of the impedance tensor and real and imaginary parts of the Wiese-vector.

A maximum of 20 iterations was allowed for the inversion to converge. In most cases, only 5 to 10 iterations were needed to reach the best fit. The fit is given as a simple rms measure of the misfit weighted by the inverse standard errors of each measurement. Thus, an rms of 1 means that data are fitted to within 1% of the impedance.

Results

The resulting models derived from inverting TE-, TM-, TE+TM-, and DET-mode data are summarized in Figure 5. The black rectangle shows the true position of the deeper 2D conductor, which has a resistivity of 2 ohm-m, and which can be thought of as the main target of the simulated geophysical survey. Profile 250 crosses the bigger circular surface conductor in the middle, and the electromagnetic response is expected to largely exhibit 2D behavior. Profile 400 crosses the central part of the "eight"-shaped conductor, and the electromagnetic response is expected to exhibit much stronger 3D effects, as indicated for instance, by phase skew values (not shown here) exceeding 0.3 (Bahr, 1991). Profile 600 crosses the smaller circular surface conductor in the middle, and the electromagnetic response is expected to show stronger 3D effects than for profile 400. Since the upper surface conductor is well reproduced by all modes, we concentrate our presentation on the deeper conductor.

TE models

Notice that the TE models are worse than the other models in reproducing the "measured" data. They also provide very

poor estimates of the deep conductor. The extremely high phases in the "observed" data are related to mode mixing when the surface conductor crosses the deeper conductor at nearly right angles. These high phases cannot be reproduced well by 2D models, but the inversion algorithm improves the fit by introducing unrealistic, small resistivities in the model (less than 0.01 ohm-m). Along profile 250, the depth to the conductor is well estimated, but the low-resistivity structure extends much too deep. Along profile 400, both the depth and the horizontal extent are about half of the true values, and along profile 600, the conductor appears as a very narrow and shallow structure.

TM models

The TM models give only bleak estimates of the true conductor. The resistivity is much too high, because the TM mode is rather insensitive to local conductors, as noted by Boerner et al. (1999). But the rms data fit is nearly perfect, which is a warning that a good rms data fit alone is insufficient to secure reliability for the estimated model.

TE+TM models

The TE+TM models look much like the TE models because the TE mode couples much better to the conductors than the TM mode. Notice, however, that the rms data fit is nearly the same as for the TE mode alone.

Z_{DET} models

As the Z_{DET} data is given approximately by the logarithmic average of TE- and TM-mode data, we would expect the corresponding Z_{DET} models to contain some features that represent logarithmic averages of the TE and TM models. Such an effect can be seen on both Z_{DET} models along the two profiles, in that their resistivities lie much closer to the true resistivities between the extreme resistivities predicted by the TE and TM models, respectively. The depth to the center of the conductor is somewhat overestimated, but the top of the conductor coincides quite well with the true depth. The most distorted model is found along profile 400, as expected. The resistivity level of the model is in good agreement with the true model, but the depth and horizontal position is displaced by about 25 m to the "north." The latter effect is also observed on the TE model and can be explained as the result of a similar shift of the high phases towards the north.

DISCUSSION

There are obvious limitations in using only two degrees of freedom for the determinant impedance, rather than four degrees of freedom in the TE and TM modes. One example is our simple 2D model with an isolated conductor and a resistor. The TE and TM impedances are particularly sensitive to such conductors and resistors, respectively. Their logarithmic average, i.e., the determinant impedance, has sensitivity

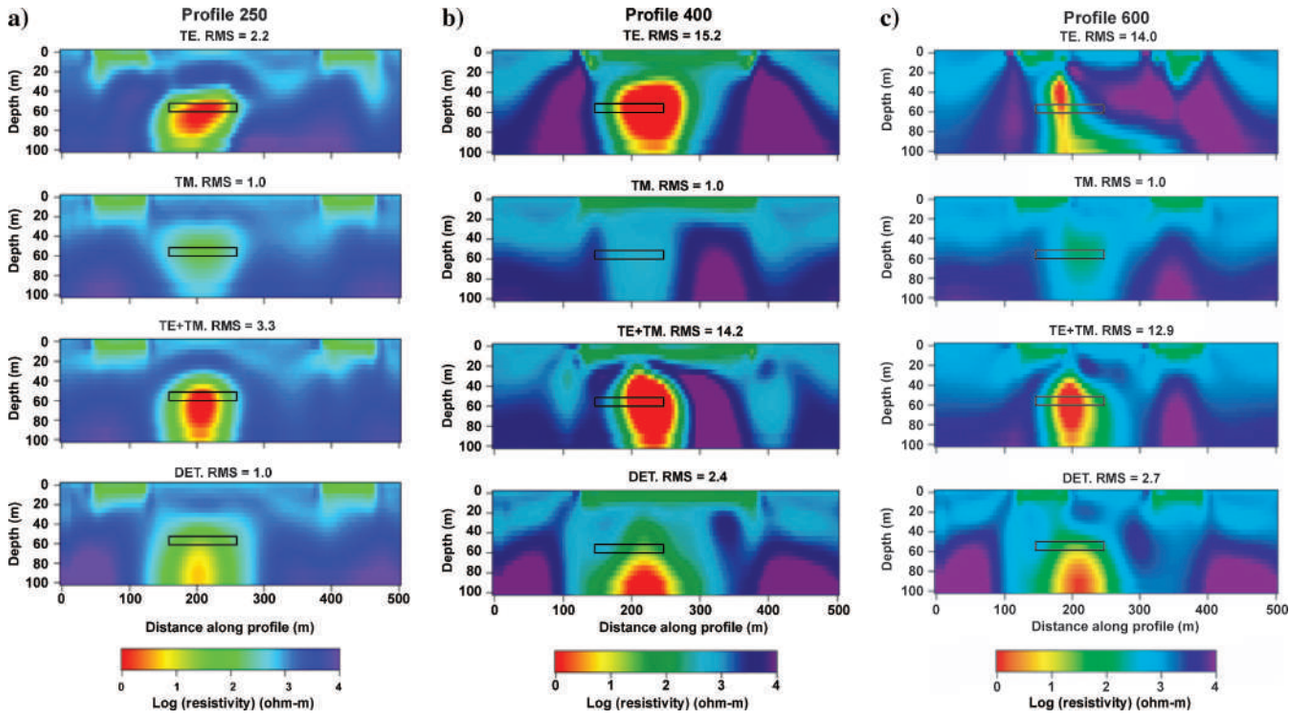


Figure 5. Inverse modeling results along profiles (a) 250, (b) 400, and (c) 600 shown in Figure 2. Initial model and a priori model are set to a homogeneous half-space of 1000 ohm-m. Black rectangles indicate the true location of the deeper conductor. Note that the estimated resistivities in the middle of the lower conductor are less than 0.01 ohm-m for TE- and TE+TM-modes along profile 400.

between the two. Therefore, if the measured data jointly satisfy 2D criteria for all stations and frequencies (McNeice and Jones, 2001), it is definitely better to simultaneously model TE and TM impedances.

If the measured data exhibit clear 3D effects, it is problematic to invert them with 2D models, irrespective of whether two, three, or four degrees of freedom are used as input data. At the present stage of development, 3D forward modeling is a standard tool for a better understanding of the most prominent conductivity structures, but it is a formidable job to quantitatively fit all eight degrees of freedom of the impedance tensor. Recently, attempts have been made to formally invert a large RMT data set with 3D models (Newman et al., 2003) using the most powerful computers. 3D inversion of MT data is a very active research area at the moment, and it will likely advance with improved algorithms and faster computers; but the normal user still must resort to 2D inverse models for the foreseeable future.

We have attempted to show that the determinant impedance can be used as a good starting point for routine 2D inversion. Although such a conclusion cannot be proven from first principles—in fact it is easy to find models for which all modes give rather poor results; for example, that of finding the fine structures in a resistive region surrounded by conductive seawater. However, the evidence presented here from the special synthetic 3D data set is very much in line with the experience we have gained over several years of determinant inversion on many artificial data sets, as well as with real data. Although not reported on in this paper, with 2D data inverted along a profile not running perpendicular to the true strike direction, it is generally possible to fit the data very well, and the distortion produced in the model is relatively small.

CONCLUSIONS

Although we cannot rule out the use of other invariants of the impedance tensor for routine inversion, we point out here some of the advantages of the determinant.

- Use of the determinant impedance is very practical because it is rotationally invariant; irrespective of the assumed strike direction used for 2D modeling, the data to be modeled is the same. Whether the estimated model represents real features, not just artifacts related to 3D effects, must be analyzed in the usual way by studying the dimensionality of the measured transfer functions.
- Determinant data can often be fitted well with 2D models (and 1D as well). We have illustrated this point with synthetic data, for which we have shown that TM models give the best fit, followed closely by determinant data, whereas TE models are much more difficult to fit.
- It becomes easy to perform automatic, routine 2D inversions without going into details concerning static shifts, change of strike direction as a function of frequency and location, and 3D effects in general. Static shifts are naturally represented by small-scale variations in the upper part of the model. Such variations produce shifts in TM-transfer functions, but the smoothness constraints provided by the regularization process prohibits these shallow and, to some extent, artificial variations to propagate to greater depths of the model.
- Responses over 3D structures often show extreme behavior for individual impedance-tensor elements, but much simpler and less distorted behavior of the determinant impedance usually can be explained well by 2D models. In our synthetic example, the TE impedance, in particular, has extreme phases approaching, and sometimes even exceeding, 90°. When attempting to reproduce such high phases with 2D inverse models, we introduced extremely low resistivities, several orders of magnitude smaller than the true ones.
- In many cases, 2D models based on the inversion of the determinant impedance arrive at reasonable approximations of real 3D structures. From our synthetic example, we see that the determinant preserves some important virtues of the TE and TM modes, while suppressing their weak sides. The TM mode is weak in sensing isolated conductors but excellent in sensing isolated resistors. For the TE mode it is vice versa. Therefore, in theory, TE and TM modes should be combined and inverted together. However, in the real 3D world, the resulting 2D models may lack the resolution that is a result of a good correspondence between the measured data and the model responses. More importantly, perhaps, they may be seriously biased as a result of mode mixing. In our experience, Z_{DET} often provides a good compromise that allows for a good data fit, and at the same time, resolving reasonably well both good and bad conducting structures along the measured profile.

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